

Lab 1

1

a

What type of register is this? (Positive Edge-Triggered Register or Negative Edge-Triggered Register). Explain and Illustrate using Sedit and simulation waveform.

✓ **Answer**

Positive edge triggered.

The first stage is negative edge triggered, but the second stage is positive edge triggered, meaning the circuit as a whole will be positive edge triggered.

When CLK is low in the first stage, D is able to open the input to $I1$ to be either V_{dd} or Gnd . When CLK is high, D is blocked from altering the output of $I1$ and instead will feed the output of $I1$ back into itself inverted, keeping the signal frozen at whatever it was before the CLK turned high.

The second stage is the opposite, allowing $I1$ to alter the output of $I2$ only if the CLK is high, and looping the output back if the CLK is low. This ultimately means the output will be whatever D was at the beginning of the positive CLK edge, before the inverters.

As seen in the graphics at the end of this document, the output only changes on positive edges of CLK .

b

Assume that the propagation delay of each clocked inverter (e.g., M1-M4) is T_{CLK_INV} and the delay of inverters $I1$ and $I2$ is T_{INV} . Derive the expression for the set-up time (T_{SETUP}) the propagation delay ($T_{CLOCK-Q}$) and the hold time (T_{HOLD}) in terms of the above parameters. Explain your results.

✓ **Answer**

T_{SETUP} must be at least long enough for the clocked inverter to propagate, then the next falling clock cycle. If at the right time, $T_{SETUP} = T_{CLK_INV}$ as any time after that, it could already be locked by the inverter.

$T_{CLOCK-Q}$ is limited only by the speed of the second clocked inverter and the final inverter, thus making its delay $T_{CLOCK-Q} = T_{CLK_INV} + T_{INV}$.

T_{HOLD} can be 0, as when it is locked after the setup time, the signal is already set and cannot change nor influence the output of that stage.

i

Measure T_{CLK_INV} , T_{INV} using Sedit and find the set-up time (T_{SETUP}), the propagation delay ($T_{CLOCK-Q}$) and the hold time (T_{HOLD}).

ii

Verify your finding by simulating the FF.

Point	Cross Number	Time (ns)
D	1	0.5000
L1	1	7.9141
I1	1	8.0084
D	2	9.5000
L1	2	9.6105
I1	2	9.6181
D	3	18.5000
L1	3	22.9145
I1	3	23.0087
D	4	27.5000
L1	4	27.6105
I1	4	27.6181
D	5	36.5000
L1	5	37.9145
I1	5	38.0087
D	6	45.5000
Q	1	45.9473
I2	1	46.0194
Qp	1	46.0555
L1	6	53.0169
I1	6	53.0318
D	7	54.5000
L1	7	54.6272
I1	7	54.7346
Q	2	90.6344
Qp	2	90.6615
I2	2	90.6883

✓ **Answer**

As seen in the graphs and table, $D2-L12$ has a delay of 0.1105ns. The delay from $L12-I12$ is 0.0076ns.

Later in the table, we can see $CLK-Q2$ has a delay of 0.1344ns, which somewhat lines up with our estimations.

We can see in the graph too that the setup time is 0, as we can switch D while swapping the clock and it will still be captured.

In the graph we can also see that we have an example of our setup time being less than 1.5ns, but it still captures, which lines up with our expectation that our setup time should be ~0.0076ns.

c

What is the function of transistors $M5-M8$ and $M13-M16$? Is this a Static CMOS circuit?

✓ **Answer**

As explained in part a, they exist to "lock" the output when the CLK is not sampling the input to that stage. It loops the output of the inverter back as the input to the clocked inverter, which keeps the output the same as whatever it was before it was locked.

Graphics

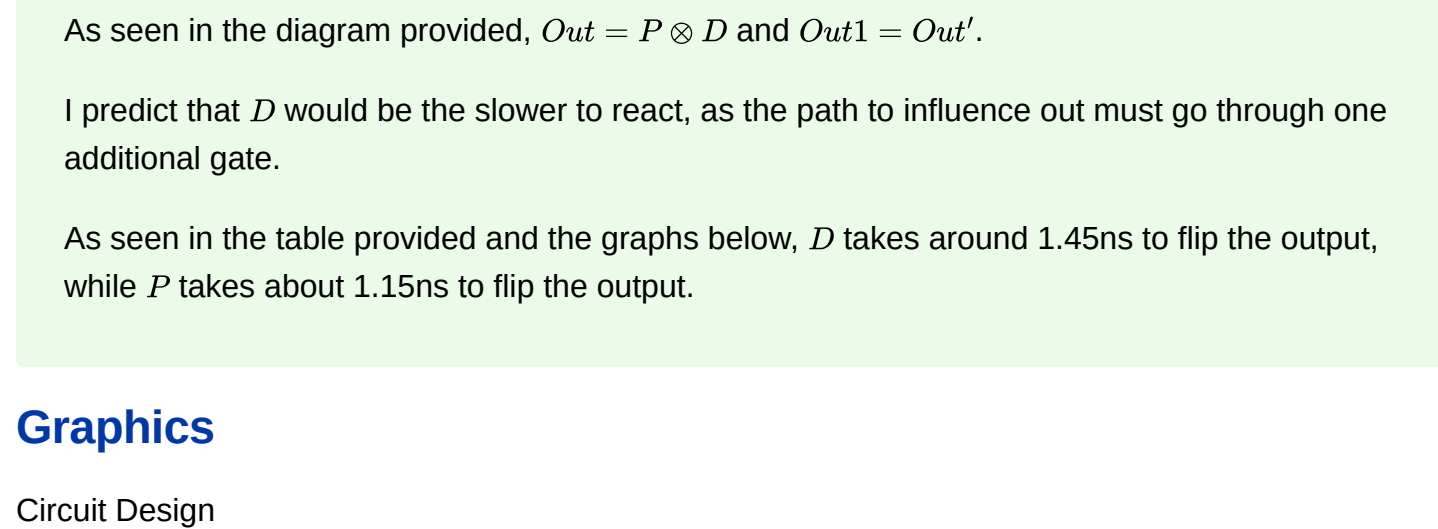
Circuit Design



Testbench Design



Waveforms



2

Using SEDIT, derive the $Out1$ and Out expressions that the circuit implements. Demonstrate the correct circuit operation and derive an input combination pairs that causes the worst-case fall time delay and worst case rise time delay at Out and $Out1$. (Assume 1ns rise and fall times for the inputs P and D).

Point	Cross Number	Time (ns)
D&P	1	0.5
D	2	15.5
Out	1	16.9308
Out1	1	16.9390
P	2	23.0
Out	2	24.1426
Out1	2	24.1577
D	3	30.5
Out	3	31.9308
Out1	3	31.9390
D&P	4&3	45.5
D	5	60.4
Out1	4	61.9308
Out	4	61.9390
P	4	68.0
Out1	5	69.1426
Out	5	69.1577
D	7	75.5
Out1	6	76.9308
Out	6	76.9390

✓ **Answer**

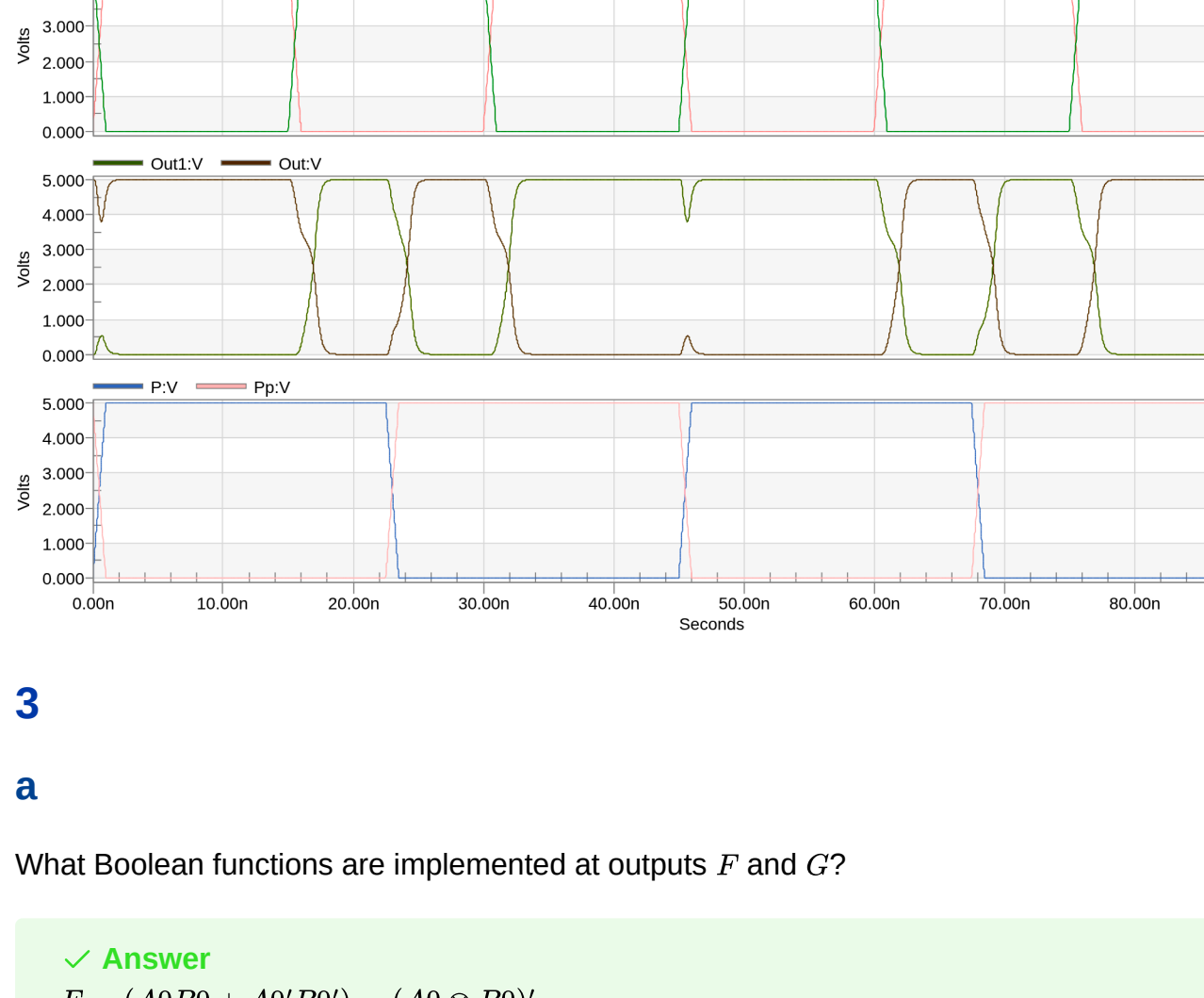
As seen in the diagram provided, $Out = P \otimes D$ and $Out1 = Out'$.

I predict that D would be the slower to react, as the path to influence out must go through one additional gate.

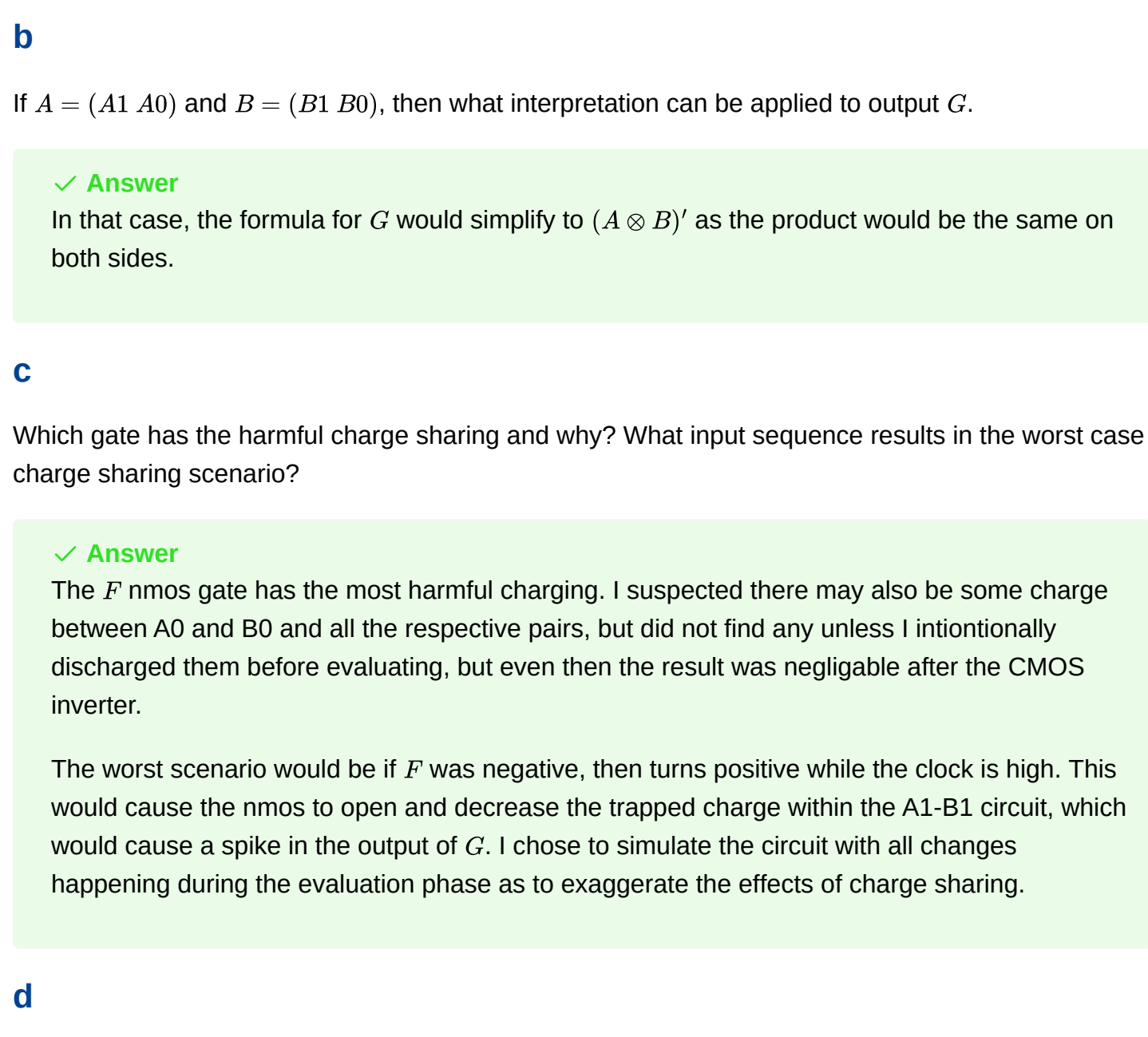
As seen in the table provided and the graphs below, D takes around 1.45ns to flip the output, while P takes about 1.15ns to flip the output.

Graphics

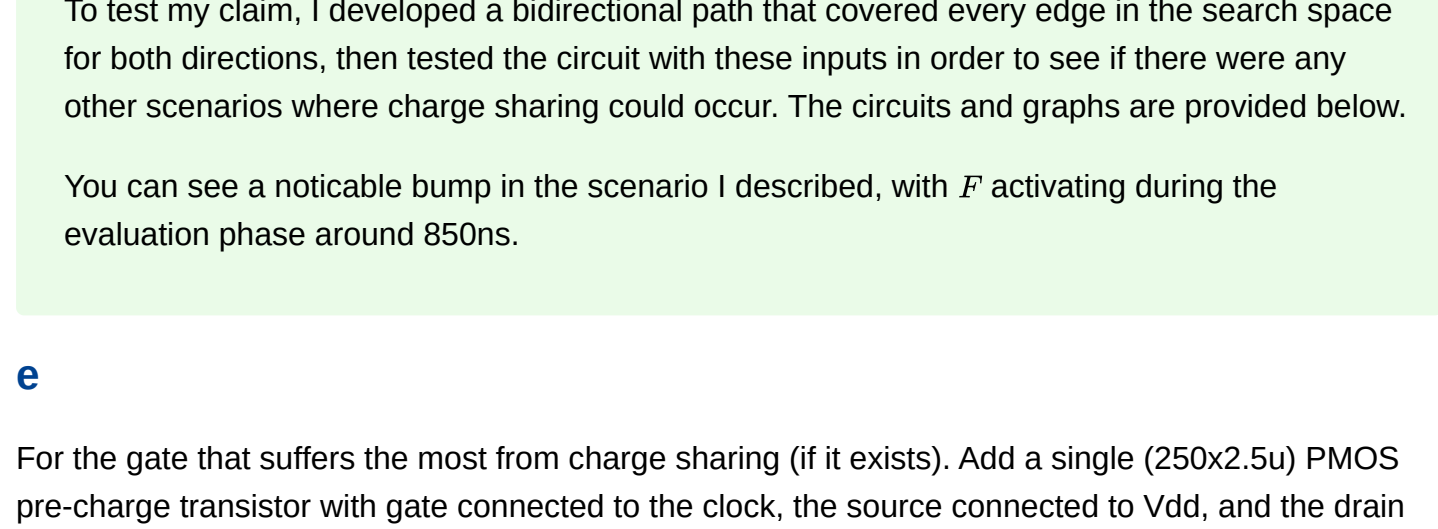
Circuit Design



Testbench Design



Waveforms



3

a

What Boolean functions are implemented at outputs F' and G' ?

✓ **Answer**

$$F = (A0B0 + A0'B0') = (A0 \otimes B0)'$$
$$G = F(A1B1 + A1'B1') = F(A1 \otimes B1)' = (A0 \otimes B0)'(A1 \otimes B1)'$$

b

If $A = (A1 A0)$ and $B = (B1 B0)$, then what interpretation can be applied to output G .

✓ **Answer**

The graphs provided below show the removal of the bumps in the scenario described, thus this solution fixes the charge sharing problem.

c

Which gate has the harmful charge sharing and why? What input sequence results in the worst case charge sharing scenario?

✓ **Answer**

The F' nmos gate has the most harmful charging. I suspected there may also be some charge between $A0$ and $B0$ and all the respective pairs, but did not find any unless I intentionally discharged them before evaluating, but even then the result was negligible after the CMOS inverter.

The worst scenario would be if F' was negative, then turns positive while the clock is high. This would cause the F' nmos to open and decrease the trapped charge within the $A1-B1$ circuit, which would cause a spike in the output of G . I chose to simulate the circuit with all changes happening during the evaluation phase as to exaggerate the effects of charge sharing.

d

Using Sedit to determine the charge sharing effects on the circuit for the worst case.

✓ **Answer**

To test my direction, I developed a bidirectional path that covered every edge in the search space for both directions, then tested the circuit with these inputs in order to see if there were any other scenarios where charge sharing could occur. The circuits and graphs are provided below.

You can see a noticeable bump in the scenario I described, with F' activating during the evaluation phase around 850ns.

e

For the gate that suffers the most from charge sharing (if it exists). Add a single (250x2.5u) PMOS pre-charge transistor with gate connected to the clock, the source connected to V_{dd} , and the drain connected to one of the nodes in that gate to maximally reduce the charge-sharing effect.

✓ **Answer**

Since the area the F' nmos was opening up causes issues when it is not pre-charged properly, I added a pmos to that area to V_{dd} to adequately charge it before evaluation time.

f

Use SEDIT to show that the transistor eliminated the charge sharing problem

✓ **Answer**

The graphs provided below show the removal of the bumps in the scenario described, thus this solution fixes the charge sharing problem.

g

For the modified circuit find a sequence of two inputs that results in the worst case delay through the circuit.

✓ **Answer**

I would choose $(A0, B0, A1, B1)$ to be $(1, 0, 1, 0)$, such that F is blocking, and the G network is grounded. Then I would transition to $(1, 1, 1, 1)$ as the F output would be precharged by then, needing to be dropped by F' activating the gate, which would only be possible after $B0$ has switched. G will only change after $B1$ has switched and the voltage has progress through G and the CMOS inverter after it.

h

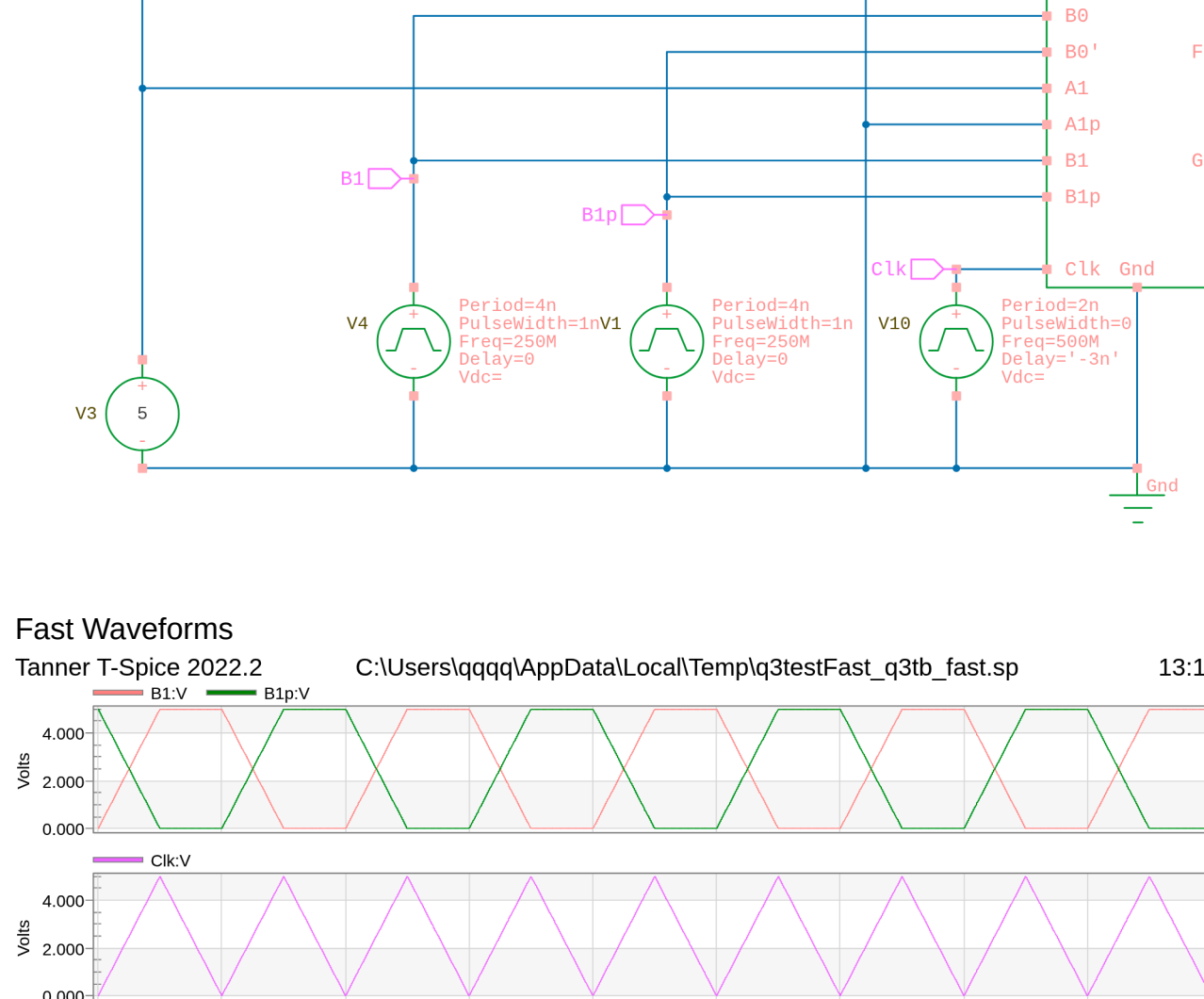
Use the simulation to find the maximum clock frequency for correct operation of the circuit.

✓ **Answer**

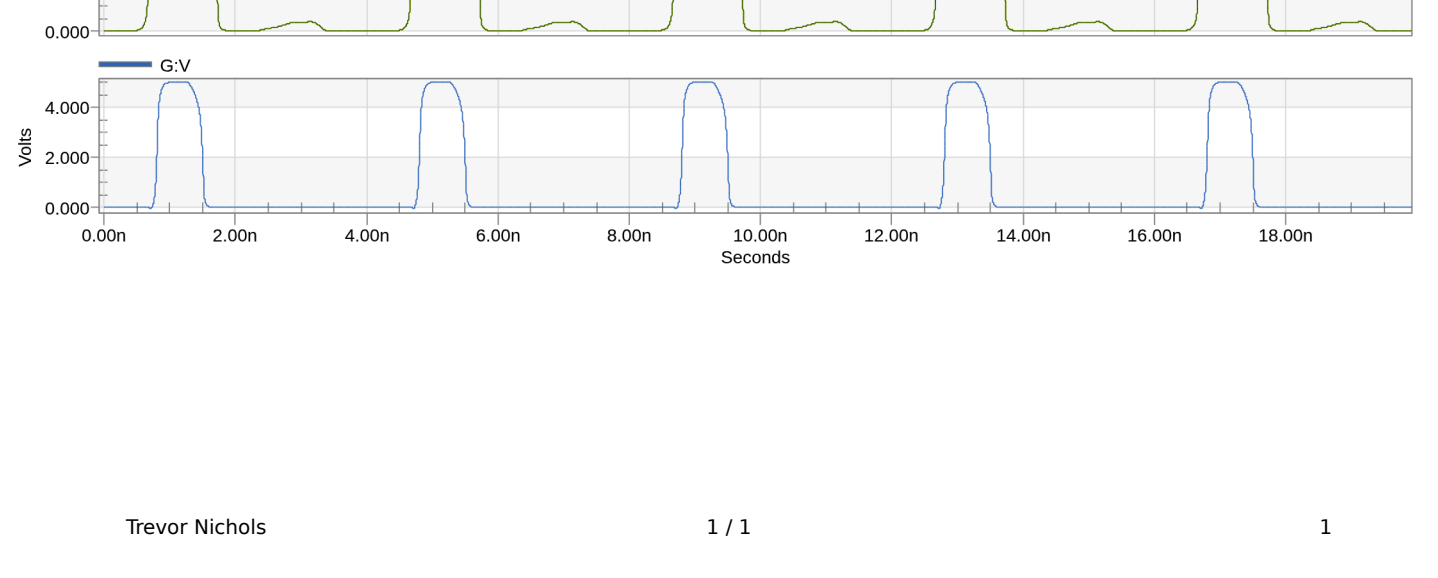
I was able to go all the way up to a clock of 2ns, which is the minimum due to the specified switching time, and the circuit still performed under the scenario I listed. Graphs are attached below.

Graphics

Circuit Design



Testbench Design



Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

Fast Waveforms

Fixed Circuit Design

Fixed Waveforms

Fast Circuit

